AI RESUME ANALYZER

Internship Project Charter

Organization

Nathcorp Private Limited

Project Title

AI Resume Analyzer

Project Manager

Somya Saraswati

Project Duration

4 Weeks

Team Size

6 Interns

Technology Stack

Frontend: React.js

Backend: FastAPI

Database: SQLite

AI Models:

- Groq API (Recommended)
- OpenRouter
- Ollama (Optional)

Version Control:
GitHub

Executive Summary

Recruiters often receive hundreds of resumes for a single position. Reviewing resumes manually is time-consuming and inconsistent.

Candidates also struggle to understand:

- Whether their resume is ATS-friendly
- Whether they match a particular job description
- Which skills they are missing
- How they should improve their resume

The purpose of this project is to build an AI-powered Resume Analyzer that helps candidates evaluate and improve their resumes using Artificial Intelligence.

The system should analyze uploaded resumes, compare them against job descriptions, calculate ATS scores, identify skill gaps, and generate personalized recommendations.

Business Objectives

The application should enable users to:

- Upload resumes
- Extract candidate information
- Analyze resume quality
- Compare resume against a job description
- Generate ATS compatibility scores
- Identify missing skills
- Recommend resume improvements
- Generate interview questions

Project Scope

In Scope

Resume Upload

Supported formats:

- PDF
- DOCX

Resume Parsing

Extract:

- Name
- Email
- Phone Number
- Skills
- Education
- Experience

ATS Score

Generate a score out of 100 based on:

- Skills
- Keywords
- Resume Completeness

Job Description Matching

Compare resume with provided JD.

Generate:

- Match Percentage
- Missing Skills
- Strength Areas

Resume Recommendations

Generate:

- Missing Keywords
- Missing Skills
- Improvement Suggestions

Interview Questions

Generate:

- Technical Questions
- Behavioral Questions
- HR Questions

Out of Scope

The following features should NOT be implemented during this internship:

- User Authentication
- Payment Integration
- Multiple User Roles
- Resume Builder
- AI Video Interviews
- Real-time Collaboration

Functional Requirements

Module 1 – Resume Upload

User uploads:

- PDF Resume
- DOCX Resume

System validates:

- File type
- File size

Expected Result:

Resume stored successfully.

Module 2 – Resume Parser

System extracts:

- Candidate Name
- Contact Details
- Skills
- Experience
- Education

Expected Result:

Structured JSON output.

Module 3 – ATS Score Calculator

System analyzes:

- Skill Coverage
- Resume Sections
- Keyword Presence

Expected Result:

ATS Score between 0-100.

Module 4 – JD Matching

Inputs:

Resume

Job Description

Outputs:

- Match Score
- Matching Skills
- Missing Skills

Module 5 – AI Recommendation Engine

Outputs:

- Resume Improvement Suggestions
- Skill Recommendations
- Learning Suggestions

Module 6 – Interview Preparation

Generate:

- Technical Questions
- HR Questions
- Scenario-Based Questions

System Architecture

React Frontend

↓

FastAPI Backend

↓

SQLite Database

↓

AI Service Layer

↓

Groq API

↓

Response Generation

Sprint Plan

Week 1

Project Setup

Tasks:

- GitHub Setup
- React Setup
- FastAPI Setup
- Upload Feature

Deliverables:

Working Resume Upload

Week 2

Resume Analysis

Tasks:

- Resume Parsing
- ATS Score Module

Deliverables:

Resume Analysis Dashboard

Week 3

AI Features

Tasks:

- JD Matching
- Recommendations
- Interview Questions

Deliverables:

Complete AI Features

Week 4

Finalization

Tasks:

- Testing
- Documentation
- Presentation

Deliverables:

Final Product

GitHub Standards

Every intern must:

- Work on feature branches
- Create Pull Requests
- Participate in code reviews
- Maintain meaningful commit messages

Example:

feature/resume-parser

feature/ats-score

feature/jd-matching

Final Deliverables

1. Source Code
2. GitHub Repository
3. Architecture Diagram
4. API Documentation
5. Setup Guide
6. Project Report
7. PowerPoint Presentation
8. Live Demonstration

Success Criteria

The project will be considered successful if:

- Resumes can be uploaded successfully
- Resume information is extracted correctly
- ATS score is generated
- JD matching works correctly
- Recommendations are meaningful
- Interview questions are generated
- Application is stable and usable